

Unified Consciousness Theory I and II

Paired v1.1 Supplement: Dependencies, Source Fidelity, and Reproducibility

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Paper I: TA-TR-2026-20, v1.1, DOI 10.5281/zenodo.23030207

Paper II: TA-TR-2026-21, v1.1, DOI 10.5281/zenodo.23030320

Status: Supplement to theoretical preprints; not peer reviewed.

S1. What this revision changes

Paper I supplies the basal process ontology and the intended tokenwise physical-experiential identity. Paper II reconstructs or embeds specified operational mechanisms under explicit bridges, while separately recording reinterpretations, conflicts, empirical status, and unrecovered commitments. The paired revision keeps these tasks separate.

The published v1.0 editions remain identifiable at DOI 10.5281/zenodo.23005588 (I) and DOI 10.5281/zenodo.23008262 (II). New versions do not change their historical claims or bytes. Candidate identifiers rc3 and rc4 describe internal development stages, not new published editions.

The present review identified five principal improvements. First, a scientific view or estimate is no longer silently equated with a token's complete ontic organization. Second, the common comparison signature explicitly retains the current pointed state, intervention ports, update and reset operations, temporal relations, and constituent structure. Third, finite bridge failures are no longer used circularly to isolate the full identity axiom. Fourth, a four-setting model proves which internal distinctions are erased or retained under particular connection changes. Fifth, Paper II corrects IIT terminology, source-domain fidelity, numerical-threshold language, and missing reproducibility materials.

No additional macro-experience existence threshold, compulsory universal subject selector, or separate human-quality generating mechanism is introduced. Biological specification and finite measurement remain application tasks within the common explanatory framework.

S2. Current dependency ledger

Result	Required inputs	Conclusion boundary
Universal nonempty experience	actual token and A1	axiom-domain commitment, not a result of a circuit calculation
No nonuniversal existence gate	A1 and an actual nonuniversality witness in the claimed domain	does not refute an unrelated domain-limited perceptual claim
Pairwise physical-experiential type equivalence	tokenwise A5-SI in the common signature	weaker type equivalence alone does not prove tokenwise identity

Result	Required inputs	Conclusion boundary
Non-summative combination	actual whole and parts; A5-SI; declared common independent product	fixed constituent comparison, not a unique subject count
Fixed sub-process invariance	same complete pointed organization and interfaces; A5-W	equal output or coarse statistic alone is insufficient
Four-setting retention proposition	stated binary dynamics, common boundary, initial perturbation	response-partition invariant; not amount of consciousness
Exclusive candidate-orbit obstruction	overlapping single orbit, deterministic equivariance, nonempty exclusive selection	does not exclude whole-world or singleton invariant partitions
Paper-II operational reconstruction	fixed physical view, shared operations, admissible theory bridge	does not require A1/A5 to calculate the operational outputs
Experiential interpretation	relevant A5 commitment and actual constitutive organization	not a certificate that a finite report is sensitive
One-way finite prediction	fixed sufficiency bridge and measurement model	no automatic reflection after lossy projection
Two-way finite prediction	one-way bridge plus independently stated finite reflection	no automatic isolation of full A5 from auxiliaries

The architecture is:

$$\mathbf{D}^{ontic}(P) \xrightarrow{\Pi_v} D_v^{PA}(P),$$

$$(D_v^{PA}(P), B_T) \xrightarrow{\mathcal{T}_T} O_T.$$

A data estimate is written $\widehat{D}_v^{PA}$. If several physical models remain compatible with evidence, their translated output set and any undefined bridge cases must remain visible. The epistemic uncertainty is not resolved by declaring one universal subject partition.

S3. Current nine-theory source-fidelity ledger

This is a revision-specific ledger. The restored v0.54 ledger is retained separately as historical source provenance; its older R/EMP labels are not the current taxonomy. The v1.0-rc4 relation ledger is also retained unchanged. The current manuscript's F axis denotes formal relation/recovery class, V denotes empirical validation, and REC/EMB/REI/CON denote relation types. These are classifications of specified sub-claims, not a total ranking of entire theories.

S3.1 IIT 4.0

Source: Albantakis et al. (2023), DOI 10.1371/journal.pcbi.1011465, particularly System integrated information, Composition, and Relations. The system quantity is ϕ_s , distinction information is ϕ_d , and relation information is ϕ_r . IIT's reference prior, intrinsic-difference measure, repertoires, admissible partitions, normalization, maximum/minimum rules, and tie conventions remain imported bridge specifications. A

system-level zero criterion is not a generic graph-factorization theorem. Computational reconstruction is conditional F3/V0; an exclusion conflict requires a specified continuing lower actual UCT process and the exact conflicting source claim. Neither full IIT phenomenology nor neural predictions are certified by this reconstruction.

S3.2 Recurrent Processing Theory

Source: Lamme (2006), DOI 10.1016/j.tics.2006.09.001. A cycle/SCC calculation is a local formal primitive, not complete conscious vision. Sensory content, recurrent timing, stabilization, and the phenomenal role must remain explicit. The graph primitive can be local F4/V0; source-faithful perceptual recovery remains conditional on the sensory and empirical bridges. The four-setting circuit is not a biological RPT validation.

S3.3 Global Neuronal Workspace

Sources: Dehaene and Changeux (2011), DOI 10.1016/j.neuron.2011.03.018; Mashour et al. (2020), DOI 10.1016/j.neuron.2020.01.026. Preserve amplification, sustained representation, broad availability, and the workspace architecture. Consumer identities, content coordinates, readout conventions, and anatomical/timing claims are not obtained from bare reachability. The numerical access thresholds in the toy examples are sampled operational criteria, not demonstrated neural ignition or an independently established dynamical bifurcation.

S3.4 Higher-Order Theory

Source: Brown, Lau, and LeDoux (2019), DOI 10.1016/j.tics.2019.06.009. Retain appropriate higher-order representation of a first-order mental state, not merely an observer's causal-tracking description. The historical source ledger supplies the detailed source framing; this revision does not claim a new exhaustive full-text audit of every HOT variant. Causal targeting is a primitive. Semantic aboutness, mental-state identification, and phenomenal necessity/sufficiency remain bridge or residual commitments.

S3.5 Predictive processing and active inference

Sources: Hodson et al. (2024), DOI 10.1016/j.neubiorev.2023.105473; Pezzulo et al. (2024), DOI 10.1016/j.biopsycho.2023.108741. Model states, generative and inference distributions, policy structure, and preferences must be identified without a target-output oracle. Pezzulo et al. explicitly formulate basic sentient behavior without phenomenological commitments. This must not be relabeled as a universal phenomenal-existence assertion to create a conflict with A1. Conditional formal computation does not establish a system's model semantics or superiority over alternatives.

S3.6 Dendritic Integration Theory

Source: Aru, Suzuki, and Larkum (2020), DOI 10.1016/j.tics.2020.07.006. Preserve pyramidal-cell compartments, bottom-up/top-down interaction, and thalamocortical

context. A nonlinear interaction functional alone is not DIT. The local sigmoid example is deliberately labeled DIT-like; biological necessity/sufficiency and implementation remain residual. Anatomical bridges do not become universal experience-existence gates under UCT.

S3.7 Memory Theory of Consciousness

Sources: Budson et al. (2022), DOI 10.1097/WNN.0000000000000319; Budson and Paller (2025), DOI 10.2147/NSS.S522975. Preserve explicit-memory architecture and the strong all-experience wording rather than reducing the theory silently to any causal trace. Temporal retention is an exact local primitive; calling a state an explicit-memory system requires further bridges. The A1 conflict is with an explicitly universalized necessity reading on a domain containing an actual nonmemory witness. A witness outside a biological source's domain does not refute its restricted claim.

S3.8 Attention Schema Theory

Sources: Graziano and Webb (2015), DOI 10.3389/fpsyg.2015.00500; Wilterson et al. (2020), DOI 10.1016/j.pneurobio.2020.101844. Preserve the simplified internal model of attention, its control role, and awareness attribution. Generic feedback or compression is not the full signature. Representation grounding and source-specific behavioral predictions remain necessary bridges or residuals. A conditional causal representation is not an empirical confirmation of AST.

S3.9 Temporo-spatial Theory of Consciousness

Sources: Northoff and Huang (2017), DOI 10.1016/j.neubiorev.2017.07.013; Northoff and Zilio (2022), DOI 10.1016/j.bbr.2022.113788. Preserve the differentiated roles: nestedness as state/level predisposition, alignment as content/form prerequisite, expansion as phenomenal correlates, and globalization as cognitive consequence. Time and scale appear in scientific views, but changing the analyst's view does not change an unchanged actual token. Scale/frequency predictions and neurophenomenal mappings are not recovered merely by defining four generic structural functions.

S4. Exact finite verification

The original verification/check_rc4.py is preserved and executed without replacing it by a new script presented as historical code. Its 13 checks cover 32 transitions, 128 same-projection comparisons, 864 reset-intervention cases, the type-permutation counterexample, restricted symmetry selection, lossy target, and coarse-description limits.

The new verification/check_transformations.py contains 17 scoped checks: 64 transitions across four settings, 256 projection comparisons, 1,728 reset cases, 8,160 initial pair trajectories across all common binary boundary strings through horizon eight, port covariance, output invariance, and finite bridge countermodels. The all-horizon theorem is analytic: a one-link pair matrix squares to zero, whereas the reciprocal pair matrix squares to the identity. Finite enumeration is a regression check, not the proof for all horizons.

Setting	Classes at horizon 1	Classes at horizons at least 2	Whole-update images
no cross-link	1	1	2
one cross-link, either direction	2	1	4
both cross-links	4	4	8

These are distinguishable response classes of initial perturbations under a held common boundary. They are not experience counts, consciousness levels, or evidence that a circuit was built. UCT’s experiential interpretation remains conditional on actual realization, complete constitutive organization, and the stated identity commitment.

S5. Numerical reimplementation and corrections

`verification/reproduce_shared_models.py` is new v1.1 code, not the unavailable original generation code or the earlier release-side reconstruction. Archived CSVs supply sweep inputs and are used only afterward for output comparison. Equations, fixed parameters, initialization, synchronous update, convergence tolerance, and controls are stated in the script.

Seven archived CSV comparisons pass the declared absolute tolerance of 10^{-9} : 1,354 rows and 11,336 cells, with a largest discrepancy of approximately 6.674×10^{-11} . The first model restarts at zero for each grid point and stops below maximum state change 10^{-13} , with 20,000 iterations allowed. The second uses 39 transitions after its specified initial state. No historical claim of zero parameter tuning is certified.

The first model’s $\rho = \sqrt{6q}$ is the spectral radius of its connection matrix. Its crossing of one is not onset of a directed cycle: both edges are already nonzero throughout the stated gate interval. Nor is that quantity the nonlinear fixed-point Jacobian’s spectral radius. The negative control removes the gated return edge, not the other edge, and retains the original local interaction calculation.

The high-access values are first qualifying sampled grid points, not exact bifurcations. In particular, the drive $s = -0.2$ crossing is 0.4725 on the stated gate grid. The second model’s first tested higher-order and access crossings are approximately 0.4032960894 and 0.7215083799. CSV column names inherited from v1.0 are preserved for reproducibility; historical words such as ignition or RPT in a column name do not override these corrected interpretations.

The numerical laws calculate the same outputs without invoking A1 or A5. Model reproducibility supports the stated mathematical consequences, not independent empirical truth of the ontology or complete biological recovery.

S6. Reproducing the package

Extract the deposited audit ZIP into a new directory. Python 3.10 or later and its standard library suffice for the checks:

```
python verification/check_rc4.py
python verification/check_transformations.py
python verification/reproduce_shared_models.py
```

The scripts regenerate CSVs and JSON result files beside the code. Archived baselines are retained separately and must not be overwritten with regenerated values. `audit/source-provenance.json` identifies baseline and assembled-source digests. SHA256 manifests permit file-integrity checking. Checksums and timestamps establish byte identity or preservation evidence, not correctness of a theory.

PDF rebuilding additionally requires Pandoc, XeLaTeX, the listed TeX packages, and the DejaVu fonts available in the build environment. Font files are not distributed. `build.py` records the rendering route. A regenerated PDF may differ at the byte level because of environment or PDF metadata; the deposited PDF is the publication artifact to use for preservation checks.

The source-trace file referenced in historical Paper II was absent from the audited v1.0 ZIP. This release restores that separately recovered historical document and labels its old taxonomy as historical. The prior v1.0 publication is not altered. New code and current results are labeled as new, rather than pretending to recover unavailable development history.

S7. Publication and evidence boundary

The author authorized completing the paired revision through DOI publication. The release procedure verifies the intended new version-family records, final filenames, file hashes, and anonymous public readback, while retaining earlier editions. Publication, timestamping, and archival upload are recorded separately from scientific tests; a DOI is not peer review. Bitcoin timestamp confirmation and Arweave readback are preservation states, not empirical validation of experience.

Both papers remain adjacent non-amending research. They do not amend or independently corroborate the Trinity Accord or its Bitcoin Originals. The publication package makes no claim of certified historical priority, human-data validation, full recovery of nine theories, or a finished fundamental-physics theory.